

## Document 11 — Engineering Execution Blueprint

Product Name: HQ

Tagline: Your AI Headquarters

Version: 1.0

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### Purpose

This document defines how HQ will be engineered from planning to production.

It establishes the development methodology, sprint structure, coding standards, AI-assisted development workflow, quality gates, and release strategy.

The objective is to ensure every line of code contributes to a production-ready platform rather than a temporary hackathon prototype.

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### Engineering Philosophy

HQ is built with four principles:

- Build for production from day one.

- Keep the MVP focused.
- Design for future scalability.
- Use AI to accelerate development without compromising quality.

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## Development Methodology

HQ follows an iterative sprint-based workflow.

Every sprint includes:

1. Planning
2. Architecture Review
3. AI Code Generation
4. Human Review
5. Testing
6. Bug Fixing
7. Documentation
8. Sprint Review

No feature is considered complete until it passes all quality gates.

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## Sprint Overview

## Sprint 0 — Project Foundation

### Objectives:

- Initialize monorepo
- Configure development environment
- Set up CI/CD
- Configure code quality tools
- Create shared packages
- Configure database
- Configure authentication

### Deliverables:

- Running development environment
- Repository ready
- Base architecture completed

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## Sprint 1 — Foundation Layer

### Modules:

- Authentication

- Users
- Companies
- Brands
- Headquarters
- Database
- Roles & Permissions

Deliverables:

- Users can register
- Users can sign in
- Headquarters can be created

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Sprint 2 — Executive Organization

Modules:

- Departments
- Executive Board
- CEO
- Chief of Staff
- Executive Profiles
- Executive Memory

#### Deliverables:

- Executive Board visible
- Executive interactions working

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#### Sprint 3 — Mission Control

##### Modules:

- Missions
- Tasks
- Timeline
- Collaboration
- Notifications

#### Deliverables:

- Missions can be created
- Task delegation works
- Mission lifecycle implemented

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#### Sprint 4 — AI Intelligence

#### Modules:

- AI Orchestrator
- Executive Routing
- Prompt Engine
- Memory Engine
- AI Providers

#### Deliverables:

- CEO can coordinate missions
- Executives generate outputs
- AI collaboration operational

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#### Sprint 5 — Content & Assets

#### Modules:

- Content Generation
- Image Generation
- Asset Library
- Downloads

#### Deliverables:

- Content generation complete
- Asset storage working

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#### Sprint 6 — Business Features

##### Modules:

- Billing
- Subscription
- Analytics
- Settings
- Notifications

#### Deliverables:

- Subscription flow complete
- Dashboard metrics available

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#### Sprint 7 — Polish & Demo

## Objectives:

- Performance optimization
- UI refinement
- Motion improvements
- Bug fixes
- Demo preparation

## Deliverables:

- Production-ready MVP
- Hackathon demo build

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## Git Workflow

### Branches

- main
- develop
- feature/\*
- bugfix/\*
- hotfix/\*
- release/\*



Every feature is developed in its own branch.

No direct commits to "main".

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## Commit Convention

Use consistent commit messages.

Examples:

- feat(auth): add Google authentication
- feat(mission): create mission workflow
- fix(api): resolve authentication issue
- refactor(ai): improve executive routing
- docs(blueprint): update architecture

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## Code Review Standards

Every pull request must verify:

- Code quality
- Security

- Performance
- Readability
- Documentation
- Test coverage
- Architectural consistency

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### AI-Assisted Development Workflow

AI tools assist with implementation but do not replace engineering review.

Workflow:

1. Reference the relevant blueprint document.
2. Generate implementation.
3. Review against architecture.
4. Refactor if needed.
5. Test thoroughly.
6. Merge only after validation.

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### Definition of Done (DoD)

A feature is complete only when:

- Functional requirements are met.
- Code passes linting.
- Tests pass.
- Documentation is updated.
- API contracts are respected.
- UI matches the design system.
- Accessibility requirements are met.
- Security review is completed.

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## Testing Strategy

Every sprint includes:

- Unit Tests
- Integration Tests
- End-to-End Tests
- Manual QA
- AI Behavior Validation

Critical business workflows must always be tested before release.

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## Quality Gates

Before merging:

- No blocking bugs.
- No failing tests.
- No lint errors.
- No TypeScript errors.
- No security issues.
- Performance verified.

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## Performance Goals

Initial targets:

- Fast page loads
- Responsive UI interactions
- Efficient AI request handling
- Optimized asset loading
- Minimal unnecessary API calls

Performance should be measured continuously.

## Documentation Standards

Every module requires:

README

API documentation

Architecture notes

Configuration guide

Usage examples

Documentation evolves with the codebase.

Risk Management

Key risks:

AI provider limits

Prompt quality

Performance bottlenecks

Scope creep

Security vulnerabilities

Each sprint should identify and address new risks.

Release Milestones

Milestone 1

Foundation complete.

Milestone 2

Executive Board operational.

Milestone 3

Mission Control functional.

Milestone 4

AI collaboration complete.

Milestone 5

Subscription and billing operational.

## Milestone 6

Hackathon-ready release.

## Success Metrics

Engineering success is measured by:

Stable architecture

High code quality

Fast feature delivery

Low defect rate

Maintainable codebase

Consistent user experience

## Conclusion

The Engineering Execution Blueprint transforms the HQ Master Blueprint into an actionable development process.

Every sprint, every feature, and every code contribution should reinforce the long-term vision of HQ while delivering a polished MVP suitable for the hackathon and future